

ELVO – Investment Portfolio & Market Analytics Platform

Introduction

ELVO is a web-based investment portfolio management platform designed to help users track their investments, monitor stock performance, and gain insights through analytics. The platform provides a centralized dashboard where users can manage their portfolios, maintain watchlists, and visualize investment growth.

The project aims to simulate the functionality of modern investment platforms while focusing on portfolio tracking and financial analytics rather than actual stock trading. It serves as a full-stack web application that demonstrates user authentication, database management, dashboard design, and data visualization.

Problem Statement

Many beginner investors find it difficult to organize and monitor their investments across multiple stocks and assets. Existing platforms often focus heavily on trading functionality, making them complex for users who simply want to track and analyze their portfolios.

ELVO addresses this problem by providing a user-friendly platform where users can:

- Track their investments in one place.
 - Monitor stock performance.
 - Maintain personalized watchlists.
 - Visualize portfolio growth through charts and analytics.
 - Access investment insights using an intuitive dashboard.
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Objectives

- Develop a secure web application for portfolio management.
 - Provide personalized investment tracking.
 - Enable users to create and manage watchlists.
 - Visualize investment data using charts and graphs.
 - Implement secure authentication and session management.
 - Offer a responsive and user-friendly interface.
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Modules to be Included

1. User Authentication Module

- User Registration
- User Login
- Logout Functionality
- Password Encryption using Bcrypt
- Session Management
- Secure Authentication

2. User Profile Module

- View Profile
- Edit Profile
- Account Information
- Portfolio Summary
- Investment History

3. Portfolio Management Module

- Add Investment
- Update Investment
- Delete Investment
- View Portfolio Holdings
- Calculate Portfolio Value

4. Watchlist Module

- Add Stocks to Watchlist
- Remove Stocks from Watchlist
- Track Favorite Stocks
- Quick Portfolio Reference

5. Dashboard Module

- Portfolio Overview
- Total Investment Value
- Profit/Loss Tracking
- Recent Activities
- Quick Statistics

6. Analytics Module

- Investment Performance Charts
- Portfolio Distribution Graphs
- Growth Analysis
- Asset Allocation Visualization

7. Search Module

- Search Stocks
- Filter Portfolio Assets
- Search Watchlist Items

8. Admin Module (Future Scope)

- User Management
 - Platform Monitoring
 - Data Analytics
 - User Activity Tracking
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Technology Stack

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- EJS Templates

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

Authentication

- BcryptJS
- Express Session
- Connect Mongo

Development Tools

- VS Code
- Git
- GitHub
- MongoDB Compass
- Postman

Deployment

- Render / Railway / Vercel (Frontend)
 - MongoDB Atlas
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System Architecture

Frontend Layer

Responsible for:

- User Interface
- Forms and User Interaction
- Dashboard Rendering
- Data Visualization

Backend Layer

Responsible for:

- API Handling
- Business Logic
- Authentication
- Database Operations

Database Layer

Responsible for:

- User Data Storage
 - Portfolio Records
 - Watchlist Information
 - Session Storage
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Workflow

User Registration

1. User opens registration page.
2. User enters details.
3. Password is encrypted.
4. User information is stored in MongoDB.
5. User is redirected to login page.

User Login

1. User enters credentials.
2. Credentials are verified.
3. Session is created.
4. User is redirected to dashboard.

Portfolio Management

1. User adds investment details.
2. Data is stored in MongoDB.
3. Dashboard updates portfolio summary.
4. Analytics module generates insights.

Watchlist Management

1. User searches stock.
 2. Stock is added to watchlist.
 3. Watchlist is displayed on dashboard.
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Future Enhancements

- Real-time Stock Market API Integration
 - AI-based Investment Recommendations
 - Portfolio Risk Analysis
 - Email Notifications
 - Mobile Application
 - Advanced Financial Analytics
 - Multi-Asset Support
 - Real-Time Market Dashboard
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Conclusion

ELVO is a full-stack investment portfolio and analytics platform developed to help users manage and monitor their investments efficiently. By integrating authentication, portfolio tracking, watchlists, analytics, and dashboard visualization, the platform provides a comprehensive investment management experience while demonstrating modern web development practices.